

Paper II — Consolidated Working Document

Statistical Organization of Localized Residual Structure in SPARC Rotation Curves

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Part I — Manuscript

Statistical organization of localized residual structure in SPARC rotation curves

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Document organization

Section	Status	Notes
Abstract	Revised v2, tightened	~250 words
§1 Introduction	First draft	Field framing, SPARC context, Paper I bridge, scope deferrals, outline
§2 Data and methods	First draft	
§3.1 Statistical organization	v2 (audit-aligned)	NGC 6674 included
§3.2 Spatial-coherence nulls	First draft	Distinguishes from Paper I nulls
§3.3 Robustness against artifacts	First draft, §3.3.5 v2	NGC 6674-included antiwarp
§4 Discussion	First draft	Constrains classes of explanation; flags backbone absorption as future work
§5 Conclusions	First draft	Statistical-not-ontological framing
Bibliography	Pending	Citation hooks in place
Acknowledgments	Pending	
Figures (10 total)	Pending; placeholders annotated in §3	Data available; generation scripts to be written

Abstract

We investigate whether the localized residual structures identified in the v7.0 SPARC rotation-curve framework of Paper I represent a statistically organized population or an accumulation of fitting artifacts. Using 102 SPARC galaxies ($T = 2-9$), the Burkert-backbone-plus-Gaussian-shell framework selects 52 shell-bearing systems via Bayesian Information Criterion, with shell-bearing fraction decreasing monotonically toward later morphological type (Spearman $\rho = -0.833$, $p = 0.010$).

The 69 fitted shells exhibit several forms of internal organization. Shell width scales approximately linearly with radius ($\sigma \propto r^{1.04}$; $\rho = +0.78$, $p < 10^{-4}$), implying a characteristic fractional width $\sigma/r \approx 0.27$ across more than two decades in radius. Within two-shell galaxies, outer shells are systematically more massive than inner shells ($p = 0.017$). Bulge-dominated galaxies are preferentially shell-bearing relative to bulgeless systems (odds ratio = 3.88, Fisher $p = 0.003$).

We test whether these patterns can arise from random or coherence-destroyed structure using two destructive null procedures applied directly to the observed rotation curves. Residual scrambling suppresses the morphology gradient, while cross-radius velocity permutation over-produces shell detections, yielding opposite failure modes relative to the real data. Shell selections additionally remain stable under perturbations to stellar mass-to-light ratio, galaxy distance, and inclination, and the principal scaling relations persist in a conservatively defined anti-warp clean subsample.

These results disfavor random fitting behavior, baryonic measurement systematics, and outer-disk warp artifacts as dominant explanations for the shell population. Whether the detected structures correspond to localized halo features or to other forms of organized galactic dynamics remains an open question.

1. Introduction

Galaxy rotation curves are commonly modeled using smooth dark-matter halo profiles such as the Navarro-Frenk-White (NFW), Burkert, Einasto, and related parameterizations. While these profiles capture the large-scale structure of many systems, observed rotation curves frequently exhibit localized residual structure relative to smooth fits. Interpreting such residuals is difficult because multiple effects can contribute simultaneously, including baryonic modeling uncertainty, noncircular motions, warped outer disks, beam-smearing effects, and fitting flexibility within the adopted halo parameterization. Distinguishing statistically meaningful structure from fitting artifacts therefore remains a central challenge in rotation-curve analysis.

The Spitzer Photometry and Accurate Rotation Curves (SPARC) database [Lelli, McGaugh, & Schombert 2016] provides a useful environment for such tests because it

combines homogeneous near-infrared photometry with high-quality HI and H α rotation curves across a broad range of galaxy morphologies. The SPARC sample has consequently become a standard benchmark for studies of galaxy dynamics, baryonic scaling relations, and halo-profile phenomenology.

Paper I introduced the v7.0 Burkert-backbone-plus-Gaussian-shell framework and showed that Bayesian model selection frequently favors localized residual components in SPARC rotation curves, with shell-bearing fraction varying systematically across morphological type. However, the existence of BIC-selected localized components alone does not establish that the detected structures correspond to physically meaningful galactic substructure. Such detections could arise from residual fitting flexibility, baryonic systematics, warp-related kinematic artifacts, or other forms of organized but nonphysical residual behavior.

The present work addresses a narrower and more fundamental question: whether the shell population identified in Paper I behaves statistically like a coherent organized population or like an accumulation of unrelated fitting artifacts. We therefore test the internal organization, spatial-coherence dependence, and robustness of the shell population under destructive null procedures and systematic perturbations, while deliberately deferring physical interpretation. The goal of the present paper is not to establish a unique physical origin for the localized residual components, but rather to determine whether the detected population exhibits statistical structure inconsistent with several major artifact channels.

This question connects to several broader areas of current research. Residual structure in galaxy rotation curves has long been discussed in the context of baryonic substructure, non-axisymmetric dynamics, and departures from idealized smooth-halo descriptions. Independent searches for dark-matter substructure have also been pursued through strong gravitational lensing, stellar-stream perturbations, and dynamical analyses of galactic halos. At the same time, modern halo modeling increasingly explores phenomenological profile families extending beyond canonical NFW forms, including cored, generalized, and feedback-modified profiles. The present work does not attempt to distinguish among these broader physical interpretations. Instead, it focuses specifically on whether the residual structures identified by the v7.0 framework exhibit reproducible statistical organization within the SPARC sample.

Several important questions are intentionally deferred beyond the scope of the present paper. We do not attempt to establish whether the localized structures correspond to spherical shells, ring-like features, or other geometric configurations, nor do we attempt an exhaustive comparison against all possible smooth-halo parameterizations or nonparametric decompositions. These questions require dedicated modeling beyond the present statistical analysis and are left for future work.

This paper is organized as follows. Section 2 summarizes the SPARC sample, the v7.0

fitting framework, and the statistical methods used throughout the analysis. Section 3 presents the empirical results in three parts: internal organization of the shell population (§3.1), spatial-coherence null tests (§3.2), and robustness against several artifact channels (§3.3). Section 4 discusses the classes of explanation constrained by these results and the principal limitations of the present analysis. Section 5 summarizes the main conclusions.

2. Data and methods

2.1 The SPARC sample and quality criteria

We work with the 102-galaxy subset of the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [Lelli, McGaugh, & Schombert 2016] as adopted in Paper I, restricted to morphological types $T = 2-9$ with quality flag $Q \leq 2$ and at least five rotation-curve points where the dark-matter contribution is positive (i.e., $V_{\text{obs}}^2 > V_{\text{bar}}^2$). The selection criteria match Paper I exactly. Each galaxy provides:

- A measured rotation curve $V_{\text{obs}}(r)$ with uncertainties $e_{V_{\text{obs}}}(r)$ at sampled radii r .
- Decomposed baryonic rotational components $V_{\text{gas}}(r)$, $V_{\text{disk}}^{\text{obs}}(r)$, $V_{\text{bulge}}(r)$, derived from the SPARC photometric and HI data.
- Catalog metadata: distance D with uncertainty e_D , inclination i with uncertainty e_i , HI extent R_{HI} , optical disk scale length R_{disk} , effective radius R_{eff} , morphological type T , and SPARC quality flag Q .
- Derived virial-scale estimates r_{vir} , halo mass M_{halo} , and stellar mass M_* via the abundance-matching procedure described in Paper I §2.1.

The sample composition by morphological type is: $T = 2$ (9 galaxies), $T = 3$ (11), $T = 4$ (17), $T = 5$ (13), $T = 6$ (16), $T = 7$ (14), $T = 8$ (6), $T = 9$ (16). Bulge presence is classified per galaxy from the SPARC photometric decomposition, yielding 24 bulge-dominated and 78 bulgeless galaxies.

2.2 The v7.0 fitting framework

We use the v7.0 framework derived and validated in Paper I. The total dark-matter contribution to the rotation curve is modeled as a Burkert backbone plus zero, one, or two Gaussian-mass shells:

$$V_{\text{DM}}^2(r) = V_{\text{Burkert}}^2(r; \rho_0, a) + \sum_{i=1}^N V_{\text{shell},i}^2(r; M_i, r_i, \sigma_i),$$

where $N \in \{0, 1, 2\}$ is selected per-galaxy by Bayesian Information Criterion. The Burkert profile has central density ρ_0 and core radius a , with enclosed mass

$$M_{\text{Burkert}}(< r) = \pi \rho_0 a^3 [\ln((1+x)^2(1+x^2)) - 2 \arctan(x)], \quad x = r/a.$$

Each shell is a localized Gaussian mass distribution with mass M_i , central radius r_i , and width σ_i . The mass enclosed within radius r is

$$M_{\text{shell}}(< r; M_i, r_i, \sigma_i) = \frac{1}{2} M_i \left[\text{erf}\left(\frac{r-r_i}{\sigma_i \sqrt{2}}\right) - \text{erf}\left(\frac{-r_i}{\sigma_i \sqrt{2}}\right) \right],$$

with $V_{\text{shell},i}^2(r) = GM_{\text{shell}}(< r)/r$.

Width constraint via reparameterization. A defining feature of the v7.0 framework is strict enforcement of the localization criterion $\sigma_i/r_i \leq 0.4$. Each shell’s width is parameterized as $\sigma_i = f_i r_i$ with f_i as a box-bounded fit parameter $f_i \in [0.01, 0.4]$, ensuring the constraint is respected at every step of the optimization regardless of optimizer trajectory. This is the principal innovation of v7.0 relative to v6.5; see Paper I §2.2 and Appendix A.

Mass bounds. Shell masses are bounded $10^6 \leq M_i \leq 5 \times 10^{10} M_\odot$. The lower bound prevents numerical degeneracies; the upper bound was chosen in Paper I to keep shells in a “localized” regime well below typical halo masses. Of the 69 shells in the v7.0 fits, 10 (14.5%) lie within 0.05 dex of the upper bound. We verify in §3.1.2 that removing these mass-bound-approaching shells produces only modest changes to the population-level scaling relations; the bound is binding for a subset of fits but does not drive the principal results.

BIC selection. For each galaxy, we fit the framework at $N = 0$, $N = 1$, and $N = 2$ shells (each with appropriate multi-restart procedure, see Paper I §2.3) and select the minimum-BIC solution:

$$\text{BIC}(N) = \chi_{\min}^2(N) + p(N) \ln(n_{\text{pts}}),$$

with parameter count $p(N) = 2 + 3N$ and n_{pts} the number of fit points after the $V_{\text{obs}}^2 > V_{\text{bar}}^2$ cut.

Baryonic decomposition. The total baryonic rotation contribution at each radius is

$$V_{\text{bar}}^2(r) = V_{\text{gas}}(r)|V_{\text{gas}}(r)| + \Upsilon_{\text{disk}} V_{\text{disk}}(r)|V_{\text{disk}}(r)| + \Upsilon_{\text{bulge}} V_{\text{bulge}}(r)|V_{\text{bulge}}(r)|,$$

with absolute values handling the SPARC sign convention for negative-contribution outer regions. Canonical mass-to-light ratios at 3.6 μm are $\Upsilon_{\text{disk}} = 0.5$ and $\Upsilon_{\text{bulge}} = 0.7$.

2.3 Sample partitions used in §3

We use three sample partitions, each suited to a specific test:

The full v7.0 sample (102 galaxies, 69 shells across 52 shell-bearing galaxies) is the canonical Paper I-aligned working set. The morphology gradient ($\rho_{\text{per-T}} = -0.833$) and most §3.1 scaling relations are computed on this sample. NGC 6674 is included as shell-bearing despite a degenerate fit ($r_1 = r_2 = 3.12$ kpc) for consistency with Paper I.

The anti-warp clean subsample (26 shells across 26 galaxies) restricts to shells satisfying four warp-protection criteria simultaneously: $r_{\text{shell}}/R_{\text{HI}} < 0.3$, disk-dominated baryonic regime at the shell radius, SPARC quality flag $Q = 1$, and inner shell of any two-shell pair. §3.3.5 uses this subsample to test whether headline patterns survive in conditions where HI-warp artifacts are most disfavored.

The two-shell paired subsample (17 paired inner-vs-outer shells from 17 two-shell galaxies) is used in §3.1.4 for paired Wilcoxon tests on within-galaxy mass and width orderings.

2.4 Methods specific to §3

The tests in §3 fall into three methodological classes:

Statistical organization tests (§3.1). Population-level Spearman rank correlations on (σ, r) , (M, r) , and σ/r -vs- r across the full shell sample. Fisher exact contingency tests on bulge presence vs shell-bearing. Paired Wilcoxon signed-rank tests on inner-vs-outer mass and width within two-shell galaxies. Two-sample Kolmogorov-Smirnov tests on r/r_{vir} distributions for inner vs outer shell populations.

Spatial-coherence null tests (§3.2). Two destructive operations on real rotation-curve data — *scramble* (within-galaxy permutation of dark-matter residuals around the canonical Burkert backbone) and *permute* (within-galaxy permutation of V_{obs} values across radii) — applied independently to each galaxy. Each null type runs 20 realizations, with the v7.0 framework applied identically to each perturbed galaxy. Test statistics are the per-T-bin and per-galaxy Spearman correlations on the resulting shell-bearing distribution, compared against the real-data baselines via empirical p -values and standardized z -scores against the null distribution.

Artifact-channel robustness tests (§3.3). Six independent tests address potential non-physical origins of the shell signal: (i) coincidence with disk dynamical scales, tested via dimensionless distance to four candidate scales; (ii-iv) systematic perturbation tests on $\Upsilon_{\text{disk}}/\Upsilon_{\text{bulge}}$, distance D , and inclination i , drawn from realistic per-galaxy uncertainty distributions and applied 20 realizations per channel; (v) the anti-warp clean subsample analysis described above; (vi) the spatial-coherence nulls of §3.2, summarized for completeness.

The perturbation tests in (ii)–(iv) use the v7.0 production fitting code (`run_canonical_fits.py`) applied to perturbed inputs, ensuring methodological identity with the canonical fits. The anti-warp filter operates on the canonical fit catalog (no refitting required). The disk-dynamical-scale test (i) uses a uniform-in-log- r null with KS-statistic comparison.

2.5 Statistical conventions

We adopt the following conventions throughout:

- Reported p -values are two-sided unless explicitly noted as one-sided.
- Spearman rank correlations are quoted with the corresponding asymptotic p -value.
- For null distributions estimated from $N = 20$ realizations, the empirical one-sided p -value floor is $1/20 = 0.05$; we additionally report standardized z -scores against the null mean and standard deviation, which are not floor-limited.
- Fisher exact tests are used for 2×2 contingency tables (e.g., bulge correlation).
- Wilcoxon signed-rank tests are used for paired within-galaxy comparisons.
- Kolmogorov-Smirnov two-sample tests are used for cumulative-distribution comparisons (e.g., inner vs outer r/r_{vir}).
- 1σ uncertainties are quoted as standard errors unless noted as bootstrap or Bayesian credible intervals.

3. Results

3.1 Statistical organization of the shell population

Paper I established that the v7.0 framework selects $n_{\text{shells}} \in \{0, 1, 2\}$ for each of 102 SPARC galaxies via Bayesian Information Criterion, with strict $\sigma/r \leq 0.4$ enforcement on shell width, yielding 52 shell-bearing galaxies (51.0%). Paper I further established the principal morphology trend: shell-bearing rate decreases monotonically with morphological lateness across $T = 2\text{--}9$, with Spearman $\rho_{\text{per-T}} = -0.833$ ($p = 0.010$) and per-galaxy $\rho_{\text{per-galaxy}} = -0.296$ ($p = 0.003$, two-sided permutation test on T-type labels).

This section examines the *internal organization* of the shell population. The morphology gradient identified in Paper I points to systematic variation between galaxies; we now ask what systematic structure exists within the 69 shells across the 52 shell-bearing galaxies. We organize the analysis around three classes of empirical pattern: (i) the bulge correlation, in which bulge-dominated galaxies are preferentially shell-bearing; (ii) the radial scaling relations, in which shell width σ and mass M scale with shell radius r ; and (iii) the geometric gradient, in which the dimensionless width σ/r varies with radius across the population.

3.1.1 Bulge correlation Bulge-dominated galaxies in the v7.0 sample are preferentially shell-bearing relative to bulgeless galaxies. Of 24 galaxies classified as bulge-dominated, 18 (75.0%) are shell-bearing. Of 78 galaxies classified as bulgeless, 34 (43.6%) are shell-bearing. The two categories together comprise the full $T = 2$ -9 sample of 102 galaxies; the SPARC photometric decomposition does not assign intermediate cases.

The shell-bearing odds ratio is

$$\text{OR} = \frac{18 \times 44}{6 \times 34} = 3.88,$$

with a Fisher’s exact test one-sided $p = 0.0028$. Bulged galaxies are $1.72\times$ more likely to be shell-bearing than bulgeless galaxies in absolute rate, equivalent to a 31.4 percentage-point difference. The bulge correlation persists in the anti-warp clean subsample (§3.3.5), with OR reduced to 2.77 due to sample-size attrition under the conservative cuts.

The bulge correlation and the morphology gradient established in Paper I are not independent: bulged galaxies are concentrated at early T-types in SPARC, so part of the bulge correlation reflects the morphology gradient and vice versa. We do not attempt to disentangle these effects in the present paper; both are reported as observational facts of the shell-bearing population.

[Figure 3.1.1 — Bulge correlation] 2×2 contingency or stacked bar showing bulged vs bulgeless shell-bearing fractions (18/24 vs 34/78). *Data source:* data/galaxy_classifications.csv (is_bulge_dom, is_bulgeless) joined with Paper I canonical CSV (shell-bearing flag). *Status:* Placeholder; generation script pending.

3.1.2 Radial scaling: width and mass The 69 fitted shells exhibit organized scaling behavior in the (σ, r) and (M, r) planes. Figure 3.1.2 shows both relations on log-log axes.

Shell width scales approximately linearly with shell radius:

$$\log_{10}(\sigma_{\text{shell}}/\text{kpc}) = (1.04 \pm 0.10) \cdot \log_{10}(r_{\text{shell}}/\text{kpc}) + (\text{TBD})$$

with Spearman rank correlation $\rho = +0.78$ ($p < 10^{-4}$, $n = 69$). Shell mass scales sub-linearly with shell radius:

$$\log_{10}(M_{\text{shell}}/M_{\odot}) = (0.76 \pm 0.14) \cdot \log_{10}(r_{\text{shell}}/\text{kpc}) + (\text{TBD})$$

with $\rho = +0.59$ ($p < 10^{-4}$, $n = 69$).

The width-radius slope close to unity ($\alpha \approx 1.04$) implies a near-universal dimensionless width σ/r : as shells span more than two decades in radius (0.2 kpc to 12 kpc), their physical widths track radius almost proportionally. The empirical median σ/r across the 69-shell sample is 0.268, with standard deviation 0.116. This dimensionless width is well below the framework cap at $\sigma/r = 0.4$ for the population center, indicating the constraint is not driving the result for typical shells.

The mass-radius slope of 0.76 indicates that more distant shells carry more enclosed mass on average, but with a sub-linear scaling. We verify the bound-state of the underlying mass parameter explicitly. The v7.0 framework imposes an upper bound on shell mass at $5 \times 10^{10} M_{\odot}$, and 10 of 69 shells (14.5%) lie within 0.05 dex of this bound (8 of 67 in the NGC 6674-excluded subset; NGC 6674 contributes 2 additional bound-pegged shells). Among the 59 shells whose mass parameter is more than 0.05 dex below the upper bound, the M-r slope is 0.67 ± 0.15 , the σ -r slope is 1.02 ± 0.11 , and the σ/r population median is 0.275 (essentially unchanged from the full-sample value of 0.268). The mass bound is therefore binding for a subset of fits but does not drive the population-level scaling relations. The slope reported above is computed on the full 69-shell sample for consistency with Paper I and §3.3.5; the bound-clean comparison establishes that the scaling pattern is not an artifact of the bound. We discuss the choice of mass bound and natural extensions in §4.

We do not interpret these slopes physically in the present paper; they are reported as empirical scaling relations of the v7.0 shell population.

[Figure 3.1.2 — Scaling relations] Two-panel log-log scatter: (left) σ vs r with linear fit (slope 1.04 ± 0.10); (right) M vs r with linear fit (slope 0.76 ± 0.14). Color-code by T-type or by inner/outer position. Mark mass-bound shells distinctly. *Data source:* data/antiwarp_per_shell.csv (use full sample, n=69; the CSV stores n=67 NGC 6674-excluded — regenerate with EXCLUDE_GALAXIES=[] or merge from canonical). *Status:* Placeholder; generation script pending.

3.1.3 Geometric gradient: σ/r varies with radius While the population-median σ/r is ≈ 0.27 , the dimensionless width is not strictly invariant across the radial range. Table 3.1.3 reports σ/r statistics by radial quartile across the 69-shell sample.

Quartile	r range (kpc)	n	σ/r median
Q1 (innermost)	0.2 – 2.2	18	0.354
Q2	2.3 – 4.4	17	0.204
Q3	4.5 – 7.0	17	0.286
Q4 (outermost)	7.3 – 12.0	17	0.185

The innermost quartile has $\sigma/r = 0.354$, the outermost $\sigma/r = 0.185$, a factor-of-two range. The middle two quartiles fluctuate more, reflecting the modest sample size at each bin. The trend suggests that shells at smaller radii occupy a larger fractional radial extent than shells at larger radii.

We emphasize that this is a population-level radial trend, not a property of paired inner-vs-outer shells within individual two-shell galaxies. Within the 17 two-shell galaxies, the paired Wilcoxon test on inner-vs-outer σ/r yields $p = 0.30$ (not significant at the within-galaxy paired level). The population-level quartile structure is the appropriate statement of the geometric gradient.

[Figure 3.1.3 — Geometric gradient] σ/r vs r scatter (log-x) with quartile boundaries marked as vertical lines and per-quartile median σ/r overlaid as horizontal bars. Reference line at $\sigma/r = 0.4$ (framework cap) and at population median 0.268. *Data source:* data/antiwarp_per_shell.csv (columns: r_sh_kpc, sigma_over_r). *Status:* Placeholder; generation script pending.

3.1.4 Inner-vs-outer in two-shell galaxies Of the 17 two-shell galaxies in the v7.0 sample, 14 (82%) have outer shells more massive than inner shells. The Wilcoxon signed-rank test for the paired difference $M_{\text{outer}} - M_{\text{inner}}$ yields $p = 0.017$ (one-sided test, alternative outer $>$ inner). Median outer-to-inner mass ratio is 1.52.

In radius, outer shells are at median $2.36\times$ the radius of inner shells (definitional ordering). Absolute width σ is also larger in outer shells: 12 of 17 paired galaxies have $\sigma_{\text{outer}} > \sigma_{\text{inner}}$, with Wilcoxon $p = 0.049$. This is consistent with the population-wide $\sigma \propto r^{1.04}$ scaling and outer shells being at larger r .

When shell positions are normalized to the host’s virial radius r_{vir} , the inner and outer populations within two-shell galaxies separate at marginal but measurable significance. Inner shells of two-shell galaxies cluster at $r/r_{\text{vir}} = 0.014$ (median, IQR [0.007, 0.021]), outer shells at 0.025 (IQR [0.015, 0.031]), with a two-sample Kolmogorov-Smirnov statistic $D = 0.47$, $p = 0.045$ ($n = 17$ inner, $n = 17$ outer). Single-shell galaxies cluster at intermediate $r/r_{\text{vir}} = 0.030$ (median), consistent with BIC-driven parsimony folding contributions into a single Gaussian when the data do not support two well-separated peaks.

This two-population substructure within shell-bearing galaxies is suggestive of two distinct dynamical populations (e.g., a more centrally-concentrated component plus a more extended component) but is not statistically definitive at the present sample size. We report it as an empirical observation requiring confirmation on larger samples.

3.1.5 Summary of organized structure Across four largely independent dimensions — bulge presence, radial mass-radius scaling, the σ/r geometric gradient, and

inner-vs-outer position within two-shell galaxies — the v7.0 shell population exhibits structure that varies systematically with measurable galaxy and shell properties:

- **Bulge correlation:** $OR = 3.88$, Fisher $p = 0.003$
- **Radial scaling:** $\sigma \propto r^{1.04}$ ($\rho = +0.78$, $p < 10^{-4}$), $M \propto r^{0.76}$ ($\rho = +0.59$, $p < 10^{-4}$), σ/r population median 0.268 ± 0.116
- **Geometric gradient:** σ/r varies from 0.354 (innermost quartile) to 0.185 (outermost quartile), a factor of ≈ 1.9
- **Inner-vs-outer in two-shell galaxies:** outer shells more massive (14/17, $p = 0.017$); outer shells broader in absolute σ (12/17, $p = 0.049$); two-population separation in r/r_{vir} at $D = 0.47$, $p = 0.045$

These patterns extend the morphology gradient established in Paper I from a between-galaxy property to an organizational property of the shell population itself. §3.2 establishes that the morphology gradient cannot be reproduced by data lacking the spatial coherence of the canonical residuals. §3.3 establishes that none of the patterns are artifacts of baryonic input systematics, warp-prone HI configurations, or coincidence with disk dynamical scales.

We do not attempt physical interpretation of these patterns in the present paper. Section 4 discusses the bounds the empirical patterns place on classes of explanation; the positive identification of a physical mechanism is the subject of future work.

3.2 Tests of spatial-coherence dependence

Paper I established two complementary null tests on the v7.0 framework. The first is a synthetic-mock false-positive analysis: 346 mocks generated from smooth Burkert-truth and NFW-truth profiles plus realistic noise, fit by the same canonical pipeline, recover shell-bearing fractions of 4.0% (Burkert-truth) and 63.6% (NFW-truth). The real-data shell-bearing rate of 51.0% is recovered at factors of 6–20 above the Burkert-truth false-positive rate in every T -bin, disfavoring smooth-Burkert truth as the dominant gradient driver. The second is a per-galaxy permutation test on the morphology gradient itself: T-type labels are permuted across galaxies and the per-galaxy Spearman ρ recomputed under each permutation, yielding a two-sided $p = 0.002$ for the observed $\rho_{\text{per-galaxy}} = -0.296$.

The tests in this section address a different question. Paper I’s analyses establish that the framework does not generate spurious shells in genuinely smooth synthetic data, and that the morphology gradient is not a chance arrangement of T-type labels. Neither test addresses whether the *spatial coherence* of the localized residual structure within each rotation curve is essential to producing the observed shell population. A galaxy’s residual signal — the difference between observed rotation V_{obs} and the smooth Burkert backbone — could be coherently structured (with characteristic radii,

widths, and amplitudes that the framework discovers as shells) or it could be incoherent fluctuation (random scatter at the noise level). If the latter, the framework’s BIC procedure should still occasionally select shells from chance configurations, but the resulting shell population should not exhibit the morphology gradient or the organized scaling relations established in §3.1.

This section tests the spatial-coherence requirement directly. We define two destructive operations on the real rotation curve data — *scramble* (shuffle DM residuals across radii within each galaxy, preserving the Burkert backbone) and *permute* (shuffle observed velocities across radii within each galaxy, destroying the Burkert backbone) — and ask whether either operation reproduces the morphology gradient when the framework is applied to the destroyed data.

For each null type and each realization, we apply the v7.0 framework (Burkert + 0/1/2 Gaussian shells, BIC-selected, $\sigma/r \leq 0.4$ enforcement) to all 102 canonical galaxies, compute the same per-T and per-galaxy Spearman correlations as the real-data baseline, and ask how often the null produces ρ as negative as the real data. The runs use 20 realizations per null type.

3.2.1 Within-galaxy residual scrambling The scramble null preserves every component of canonical analysis except the spatial coherence of dark-matter residuals. For each galaxy, we compute the dark-matter residual from the canonical Burkert fit:

$$\epsilon(R) = V_{\text{obs}}^2(R) - V_{\text{bar}}^2(R) - V_{\text{Burkert}}^2(R)$$

permute the ϵ values across radii (preserving the multiset of residual amplitudes but destroying their order in R), and reconstruct a synthetic observed velocity curve:

$$V_{\text{obs}}^{\text{scramble}}(R) = \sqrt{V_{\text{bar}}^2(R) + V_{\text{Burkert}}^2(R) + \epsilon^{\text{shuffled}}(R)}$$

The scramble preserves V_{bar}^2 at each radius, the Burkert backbone at each radius, the per-radius error bars, and the radial sampling. It destroys only the within-galaxy radial coherence of localized residual structure.

Across 20 scramble realizations on 102 galaxies (2,040 fits total), the null distribution of $\rho_{\text{per T}}$ has mean -0.197 with standard deviation 0.146 , range $[-0.539, +0.095]$. The real-data baseline of -0.833 falls 4.4σ below the null mean. None of 20 realizations produces a $\rho_{\text{per T}}$ as negative as real data; the empirical one-sided p -value is $p_{\text{emp}} < 0.05$ (0/20, limited by realization count). The per-galaxy correlation $\rho_{\text{per galaxy}}$ has null mean -0.150 ± 0.048 , range $[-0.241, -0.095]$, with the real-data baseline of -0.296 at 3.0σ below the null mean and again 0/20 realizations matching it.

The scramble null does retain a weakly negative ρ_{perT} on average (-0.197). This is interpretable: the scramble preserves each galaxy’s smooth Burkert backbone, and Burkert backbone parameters correlate with morphological type (early-type galaxies have more concentrated halos, deeper baryonic potentials, and more complex residual structure for the framework to find by chance). A correlation of -0.197 is what residual *amplitude variations* alone — without spatial coherence — can produce. The real-data correlation of -0.833 is more than four times stronger, demonstrating that the morphology gradient depends on the *spatial coherence* of localized residual structure, not merely its amplitude distribution.

3.2.2 Cross-radius V_{obs} permutation The permutation null breaks both the localized residual structure and the smooth backbone. For each galaxy, we shuffle V_{obs} values across radii while holding R values fixed, then refit the v7.0 framework. The permutation preserves each galaxy’s V_{obs} amplitude distribution and its radial sampling. It destroys all coherent radial structure: the smooth rotation-curve shape, the residual-from-backbone, and any localized features.

Across 20 permutation realizations on 102 galaxies, the null distribution of ρ_{perT} has mean $+0.375$ with standard deviation 0.232 , range $[-0.169, +0.790]$. **The sign of the null correlation is opposite the real-data correlation.** The real-data baseline of -0.833 falls 5.2σ below the null mean. The empirical $p < 0.05$ (exact $0/20$ as negative as real). The per-galaxy correlation has null mean $+0.162 \pm 0.055$, range $[+0.044, +0.250]$ — also positive — with the real-data baseline at 8.3σ below the null mean.

The positive null correlation in permuted data is not a defect; it is informative. When we permute V_{obs} across radii, the resulting “rotation curve” is a chaotic profile inconsistent with any smooth halo. The BIC procedure responds to this chaos by selecting shell-bearing fits at greatly elevated rates: across all T-types, mean shell-bearing fractions in permuted data are 0.72 – 0.97 , compared to 0.31 – 1.00 in real data and 0.15 – 0.79 in scrambled data (Table 3.2.1). The framework over-detects shells in permuted data because the BIC penalty is exceeded by the χ^2 improvement from any localized component fitted to chaotic noise.

The positive ρ_{perT} in permutation arises from the small population of permuted fits where the framework cannot fit shells well: these are scattered across morphological types in a pattern that yields a positive Spearman ρ by chance. The point is not the sign per se, but that the *direction* of the morphology gradient under permutation is opposite to real, with no overlap in 20 realizations.

3.2.3 Per-T shell-bearing fractions: the failure modes are physically distinct

The two nulls fail to reproduce real data in opposite directions, which is itself diagnostic. Table 3.2.1 shows the per-T-bin shell-bearing fraction in real data alongside

the mean fraction across 20 null realizations.

T	N (real)	Real frac.	Scramble null	Permute null	Real / Scramble	Real / Permute
2	9	1.000	0.794	0.961	1.26	1.04
3	11	0.545	0.150	0.723	3.64	0.75
4	17	0.529	0.309	0.815	1.71	0.65
5	13	0.538	0.308	0.923	1.75	0.58
6	16	0.562	0.291	0.972	1.94	0.58
7	14	0.357	0.225	0.921	1.59	0.39
8	6	0.333	0.383	0.925	0.87	0.36
9	16	0.312	0.231	0.969	1.35	0.32

Two patterns are evident. The scramble null produces shell-bearing fractions *below* real data for most T-types (real exceeds scramble by factors of 1.26–3.64×, except T=8 where the real fraction is 0.87× the scramble null due to small-bin statistics). This is consistent with the scramble destroying the localized features that drive real-data shell detections. The permute null produces shell-bearing fractions *above* real data for nearly every T-type (real / permute ratios of 0.32–1.04). This is consistent with the permute creating chaotic profiles that the BIC procedure over-fits with shells.

Neither null reproduces real data, but they fail in opposite directions: scramble *under-detects*, permute *over-detects*. This asymmetry is incompatible with the hypothesis that real shell selections are random framework responses. A random-detection scenario would show null detection rates bracketing real rates symmetrically, not bracketing them on opposite sides.

3.2.4 Joint interpretation Both nulls reject the hypothesis that the morphology gradient could arise from random residual structure or random rotation-curve amplitudes:

- $\rho_{\text{per T}}$ rejected at $> 4\sigma$ under scramble, $> 5\sigma$ under permutation
- $\rho_{\text{per galaxy}}$ rejected at $> 3\sigma$ under scramble, $> 8\sigma$ under permutation
- Empirical $p < 0.05$ in all four tests (limited by $N = 20$ realization resolution)

The two nulls bracket a meaningful range of artifact channels: scramble tests whether radial coherence within each galaxy is necessary; permutation tests whether *any* coherent rotation-curve shape is necessary. The morphology gradient in v7.0 shell selections survives in neither. The two failure modes — scramble under-detecting because spatial coherence has been destroyed, permute over-detecting because the BIC procedure fits chaos — establish that the real-data shell population is not reproducible by fitting to coherence-destroyed residuals or fully randomized rotation curves.

[Figure 3.2.1 — Null distributions of ρ_{perT}] Two-panel figure: (left) histogram of ρ_{perT} across 20 scramble realizations, real-data baseline of -0.833 marked as vertical line. (right) Same for permutation null. Annotate null mean $\pm 1\sigma$ and z-score of real baseline. *Data source:* data/nulltest_per_realization.csv (filter by null_type='scramble' and 'permute'; column: rho_per_T). *Status:* Placeholder; generation script pending.

[Figure 3.2.2 — Per-T shell-bearing fractions] Grouped bar chart: for each T-type (2–9), three bars side-by-side — real data, scramble null mean, permute null mean. Error bars on null means from realization std. Shows asymmetric failure modes (scramble below real, permute above). *Data source:* data/nulltest_summary.txt for real data; data/nulltest_per_realization.csv for null means (columns frac_T2 through frac_T9 averaged over realizations per null_type). *Status:* Placeholder; generation script pending.

3.3 Robustness against artifact explanations

The empirical patterns documented in §3.1 — population-wide mass-radius scaling, near-universal σ/r , and structural correlations with galaxy morphology and bulge presence — could in principle arise from sources other than physical localized mass features in galaxy halos. Six potential artifact channels are testable within the SPARC sample and the v7.0 framework: (i) coincidence with disk-baryonic dynamical scales, (ii–iv) systematic measurement uncertainty on the three principal baryonic inputs (mass-to-light ratio, distance, inclination), (v) HI disk warps producing localized line-of-sight velocity features, and (vi) random structure in the data being misclassified as shells by the BIC selection. We address each in turn. The null tests of channel (vi) appear in §3.2 and are summarized at the end of the present section for completeness.

3.3.1 Coincidence with disk-baryonic dynamical scales If the localized features detected by v7.0 originated in baryonic disk substructure rather than halo mass features, shell radii should preferentially cluster at radii associated with disk dynamical scales: the disk scale length R_{disk} , two-and-a-fifth disk scale lengths ($2.15 R_{\text{disk}}$, the canonical exponential-disk rotation peak), the half-light radius R_{eff} , or the radius of the observed rotation curve maximum. We test this by computing, for every detected shell, the dimensionless distance $|r_{\text{shell}} - r_{\text{scale}}|/r_{\text{scale}}$ for each of the four candidate scales, and comparing the observed distribution to the expectation under a uniform null where shell radii are drawn from a power-law in r within each galaxy’s fitted range.

The observed distributions show no preferential clustering at any of the four candidate scales. The Kolmogorov-Smirnov statistic against the uniform null fails to reject at the 0.05 level for all four scales. The mean and median dimensionless offsets are consistent with random placement. We conclude that shell radii are not coincident with disk dynamical features, disfavoring a simple disk-resonance or disk-substructure origin for the population-level signal.

[Figure 3.3.1 — Disk dynamical scale coincidence] Four-panel CDF comparison. Each panel: empirical CDF of $|r_{\text{shell}} - r_{\text{scale}}|/r_{\text{scale}}$ for one candidate scale ($R_{\text{disk}}, 2.15 R_{\text{disk}}, R_{\text{eff}}, r$ at $V_{\text{rot}}^{\text{peak}}$), overlaid on uniform-placement null CDF. KS p -value annotated per panel. *Data source:* Paper I canonical CSV (shell radii) + data/sparc_sample123.csv (Rdisk, Reff) + Rotmod_LTG files (V_{rot} peak radius). *Producer script* not yet written. *Status:* Placeholder; generation script pending.

3.3.2 Systematic perturbations: mass-to-light ratio Stellar mass-to-light ratios in SPARC are nominally fixed at $\Upsilon_{\text{disk}} = 0.5$ and $\Upsilon_{\text{bulge}} = 0.7$ at $3.6 \mu\text{m}$ [Lelli2016]. The systematic uncertainty on these ratios is approximately 0.1 dex. We test whether shell selections survive realistic variation in Υ by performing 20 independent perturbation realizations: for each realization we draw $(\log_{10} \Upsilon'_{\text{disk}}, \log_{10} \Upsilon'_{\text{bulge}})$ from independent normal distributions with means $(\log_{10} 0.5, \log_{10} 0.7)$ and standard deviation 0.1 dex, and refit the v7.0 framework on all 102 galaxies with the perturbed Υ values. Each refit follows the canonical v7.0 procedure: Burkert + 0/1/2 Gaussian shells, BIC-selected with parameter penalty $k = 2/5/8$ and $\sigma/r \leq 0.4$ wall.

Across 2,001 successful refits (102 galaxies $\times$ 20 realizations, with 39 fits failing on $V_{\text{obs}}^2 > V_{\text{bar}}^2$ point cuts after Υ rescaling), shell-number selections agree with the canonical v7.0 result in 86.2% of cases (1,725/2,001). At the per-galaxy level, the modal n_{shells} across the 20 realizations matches the canonical fit in 95.1% of galaxies (97/102). Stratifying by canonical shell count: galaxies with canonical $n_{\text{shells}} = 0$ show 92.3% per-fit stability, $n_{\text{shells}} = 1$ shows 83.5%, and $n_{\text{shells}} = 2$ shows 74.3%. Among galaxies whose shell count is fully stable across all 20 realizations and that contain at least one shell, the median scatter in $\log_{10} r_{\text{shell},1}$ is 0.017 dex (approximately 4% in radius).

The hierarchy is consistent with information-theoretic expectations: galaxies whose canonical fit is in the most parsimonious branch ($n_{\text{shells}} = 0$) are least likely to flip under perturbation, while those at the BIC margin between two-shell and one-shell solutions are most likely to flip across the boundary. The aggregate population fractions (n_0, n_1, n_2) shift from (49.0%, 34.3%, 16.7%) canonical to (48.2%, 35.7%, 14.2%) under perturbation — a population-level change of less than 2.5 percentage points in any bin.

The shell signal is therefore not driven by baryonic mass-to-light mismodeling.

[Figure 3.3.2 — Υ perturbation stability] Left: stacked bar of perturbed-vs-canonical n_{shells} matches across 2,001 fits, stratified by canonical shell count ($n=0$: 92.3%, $n=1$: 83.5%, $n=2$: 74.3%). Right: scatter of $\log_{10} r_{\text{shell},1}^{\text{pert}}$ versus $\log_{10} r_{\text{shell},1}^{\text{canon}}$ for fully-stable shell-bearing galaxies, $\pm 25\%$ bands marked. *Data source:* data/upsilon_perturbation_per_galaxy.csv (columns: status, n_shells, r_sh1) joined with Paper I canonical CSV. *Status:* Placeholder; generation script pending.

3.3.3 Systematic perturbations: distance Galaxy distances enter the rotation curve fit through two channels: physical radii scale as $r \rightarrow r \cdot (D'/D)$, and baryonic rotation contributions scale as $V_{\text{bar}} \rightarrow V_{\text{bar}} \cdot \sqrt{D'/D}$ (since baryonic mass scales as D^2 at fixed observed flux while $r \propto D$, giving $V^2 \propto M/r \propto D$). Observed rotation velocities V_{obs} are distance-invariant. Each SPARC galaxy carries a published distance D and a per-galaxy uncertainty e_D reflecting the measurement method (Hubble flow, TRGB, Cepheid, etc.).

We test distance-systematic robustness by 20 realizations in which each galaxy independently draws $D' = D \cdot 10^{\mathcal{N}(0, \sigma_{\log D})}$ with $\sigma_{\log D} = e_D / (D \ln 10)$, refits the v7.0 framework on perturbed-radius and perturbed-baryon data, and records shell selections. The fractional perturbation magnitudes span the published range: median 8.7%, mean 13.8%, 90th percentile 31.8%, with one outlier (a dwarf with $e_D > D$) reaching 132%.

Across 2,036 successful refits (4 fits failing on insufficient DM points), shell-number selections match canonical in 89.6% of cases. At the per-galaxy level, modal shell count matches canonical in 94.1% of galaxies. Stratification by canonical n_{shells} yields stability rates of 94.6% ($n = 0$), 86.1% ($n = 1$), and 82.3% ($n = 2$). Median $\log_{10} r_{\text{shell},1}$ scatter for fully-stable galaxies is 0.039 dex ($\sim 9\%$ in radius). Population fractions shift by less than 2.5 percentage points in any bin.

We note that distance perturbation produces *more* stable shell selections than mass-to-light perturbation in our tests. This is physically reasonable: distance scaling is shape-preserving, multiplying all radii by the same factor and all baryonic velocities by the same factor, so the relative shape of $V_{\text{DM}}^2 = V_{\text{obs}}^2 - V_{\text{bar}}^2$ is preserved. Mass-to-light perturbation, by contrast, redistributes the disk-versus-bulge contributions to V_{bar} at different radii (since V_{disk} and V_{bulge} profiles peak at different scales), producing more disruptive changes to the residual structure being fit.

The shell signal is not driven by distance measurement uncertainty.

[Figure 3.3.3 — Distance perturbation stability] Same two-panel format as Figure 3.3.2, for distance perturbation. Stability stratification: $n=0$: 94.6%, $n=1$: 86.1%, $n=2$: 82.3%. Includes per-galaxy fractional perturbation magnitude (median 8.7%). *Data source:*

data/distance_perturbation_per_galaxy.csv (columns: status, n_shells, r_sh1, distance_factor) joined with Paper I canonical CSV. *Status*: Placeholder; generation script pending.

3.3.4 Systematic perturbations: inclination Galaxy inclinations i enter rotation curve fitting through the deprojection $V_{\text{obs}} = v_{\text{los}} / \sin i$, where v_{los} is the observed line-of-sight velocity. SPARC publishes per-galaxy inclinations i with uncertainties e_i typically in the range 1° - 5° . We test inclination-systematic robustness by 20 realizations in which each galaxy independently draws $i' = i + \mathcal{N}(0, e_i)$, with i' floored at 10° and capped at 89° to avoid pathological behavior near the deprojection singularity. We rescale $V_{\text{obs}} \rightarrow V_{\text{obs}} \cdot \sin i / \sin i'$ and $e_{V_{\text{obs}}} \rightarrow e_{V_{\text{obs}}} \cdot \sin i / \sin i'$; physical radii and modeled baryonic components are inclination-invariant.

Eleven SPARC galaxies in the canonical sample are catalogued at exactly $i = 90^\circ$ (edge-on); these are excluded from the perturbation procedure to avoid asymmetric perturbations against the upper bound. The exclusion is conservative: at $i = 90^\circ$, $\sin i = 1.000$, and at $i = 89^\circ$, $\sin i = 0.9998$, so inclination perturbations at near-edge-on inclinations produce negligible V_{obs} rescaling. Including these galaxies would only increase the apparent stability rate. The final perturbation suite comprises 1,820 fits across 91 galaxies $\times$ 20 realizations.

Realized perturbation magnitudes: median $|i' - i| = 1.87^\circ$, mean 2.96° , maximum 28.8° . The implied V_{obs} rescaling factor $\sin i / \sin i'$ ranges over $[0.681, 2.359]$.

All 1,820 fits succeeded. Shell-number selections match canonical in 95.7% of fits (1,741/1,820). At the per-galaxy level, modal shell count matches canonical in 98.9% of galaxies. Stratification by canonical n_{shells} : 99.6% ($n = 0$), 91.6% ($n = 1$), 90.7% ($n = 2$). Median $\log_{10} r_{\text{shell},1}$ scatter for fully-stable galaxies is 0.005 dex ($\sim 1.2\%$ in radius) — the tightest position recovery of the three perturbation tests.

The inclination perturbation produces the most stable shell selections of the three baryonic-input systematics. This is physically reasonable: inclination perturbation rescales V_{obs} uniformly across all radii, preserving the *shape* of the rotation curve while changing only its overall amplitude. The shell-fitting machinery operates on residual shape relative to the smooth Burkert backbone, which is largely preserved under uniform amplitude rescaling.

The shell signal is not driven by inclination measurement uncertainty.

[Figure 3.3.4 — Inclination perturbation stability] Same two-panel format as Figure 3.3.2, for inclination perturbation. Stability stratification: $n=0$: 99.6%, $n=1$: 91.6%, $n=2$: 90.7%. Tightest position recovery of three perturbation tests (median $\Delta \log r_1 = 0.005$ dex). *Data source*: data/inclination_perturbation_per_galaxy.csv (91 galaxies $\times$ 20 reps;

11 edge-on excluded) joined with Paper I canonical CSV. *Status*: Placeholder; generation script pending.

3.3.5 Anti-warp clean subsample The most plausible mechanism by which a smooth halo could mimic a localized residual feature is a velocity-field artifact in the outer rotation curve — most commonly a kinematic warp where the gas disk and the inner stellar disk no longer share a common rotation axis. Warp-induced line-of-sight velocity offsets manifest as apparent residuals at large radii where neutral hydrogen dominates the kinematic tracer and where the assumed disk inclination becomes least reliable. If the v7.0 shells were dominated by warp artifacts, the headline empirical patterns of §3.1 should weaken substantially upon excluding shells in conditions most susceptible to such artifacts.

We define a “clean” anti-warp subsample as those shells satisfying *all four* of the following conditions simultaneously:

1. **Shell radius well inside the HI gas extent:** $r_{\text{shell}}/R_{\text{HI}} < 0.3$ (excludes shells in the outer-disk regime where warps preferentially appear)
2. **Disk-dominated at the shell radius:** $V_{\text{disk}}^2 + V_{\text{bulge}}^2 > V_{\text{gas}}^2$ at r_{shell} (excludes shells in gas-dominated regions where warp signatures concentrate)
3. **Highest-quality SPARC rotation curve:** $Q = 1$ flag in the SPARC catalog (excludes galaxies flagged for tilted-ring-fit irregularities)
4. **Inner shell of any two-shell pair:** drops the outer shell of $n = 2$ galaxies (the more warp-vulnerable position)

These cuts are deliberately conservative; each independently is well-motivated, and the conjunction selects shells where multiple lines of warp protection are active. Of the 69 shells in the full v7.0 sample, 26 (37.7%) survive all four cuts. Considered independently, the cuts pass 50, 67, 48, and 52 shells respectively; the combined intersection is 26.

The 26-shell anti-warp clean subsample exhibits the same headline patterns as the full population. Table 3.3.5 reports the comparison.

Quantity	Full sample (n = 69)	Clean subsample (n = 26)
M - r slope ($\log M$ vs $\log r$)	0.76	0.69
M - r Spearman ρ	+0.59 ($p < 10^{-4}$)	+0.48 ($p = 0.013$)
σ - r slope ($\log \sigma$ vs $\log r$)	1.04	0.92
σ - r Spearman ρ	+0.78 ($p < 10^{-4}$)	+0.76 ($p < 10^{-4}$)
σ/r population median	0.268	0.310
σ/r population std	0.116	0.128
Bulge correlation OR	3.88 ($p = 0.003$)	2.77 ($p = 0.022$)
Morphology gradient $\rho_{\text{per-T}}$	−0.83 ($p = 0.010$)	−0.69 ($p = 0.058$)

The five organizational signatures of §3.1 — bulge correlation, $M \propto r$ scaling, $\sigma \propto r$ scaling, σ/r universality, and the morphology gradient — all persist on the conservatively cleaned subsample. Effect sizes are modestly attenuated, which is expected given the 62% reduction in sample size and the conservative nature of the cuts (each of which removes shells whether or not they were artifacts). Effect *directions* are preserved exactly. The morphology gradient $\rho_{\text{per-T}} = -0.69$ on the clean subsample is just outside the $p < 0.05$ threshold ($p = 0.058$, $n = 8$ T-bins), driven by the small per-bin counts after attrition rather than by a directional reversal.

We emphasize what would have happened if warp artifacts dominated the shell population: at minimum, the M - r slope would have flattened substantially (warp artifacts produce shells at the warp radius, not at radii correlated with intrinsic enclosed mass), the bulge correlation would have weakened or vanished (warps are not preferentially present in bulged galaxies), and the morphology gradient would have weakened (warps occur in late-type gas-rich galaxies more frequently than early-type bulged systems, producing the opposite gradient direction to the one observed). None of these falsifiers materializes. The bulge correlation OR drops from 3.88 to 2.77 — an attenuation, not a removal. The M - r slope drops from 0.76 to 0.69 — within statistical fluctuation given the smaller sample. The morphology gradient drops from $\rho = -0.83$ to $\rho = -0.69$ — same direction, slightly softer.

A more direct demonstration of warp-versus-shell separability would require model comparison against an explicit kinematic-warp model fit to each rotation curve, which is beyond the scope of the v7.0 framework. We do not claim the present test is exhaustive. We claim only that the headline empirical patterns are not localized to the warp-prone regime where the most plausible artifact mechanism would concentrate. Combined with the §3.3.1 disk-dynamical-scale null, the §3.3.2-3.3.4 baryonic-perturbation tests, and the §3.2 spatial-coherence nulls, this places the warp-artifact hypothesis as one of several plausible alternatives that the data do not support.

[Figure 3.3.5 — Anti-warp clean subsample] Two-panel comparison: σ -vs- r scatter (left) and M -vs- r scatter (right). Full-sample shells ($n=69$) in light gray; anti-warp clean shells ($n=26$) highlighted. Linear fits overlaid for each population with slopes annotated (full vs clean: M - r 0.76->0.69, σ - r 1.04->0.92). *Data source:* data/antiwarp_per_shell.csv (use is_clean flag for the clean subsample). *Status:* Placeholder; generation script pending.

3.3.6 Random-structure null tests (cross-reference to §3.2) The scrambling and permutation null tests of §3.2 establish that the shell selections are statistically distinguishable from random structure by z -scores of -4.36 and -5.21 for $\rho_{\text{per-T}}$ under scramble and permute respectively, and -3.04 and -8.33 for $\rho_{\text{per-galaxy}}$. Both tests target the hypothesis that the BIC-selected shells could be detector-flexibility artifacts of fitting random or shape-randomized data. The asymmetric failure modes (scram-

ble under-detects, permute over-detects) further establish that the shell population is a property of spatially coherent localized residual structure, not a property that survives destruction of either the residual coherence (scramble) or the rotation-curve shape (permute).

3.3.7 Combined verdict The localized structures detected by v7.0 satisfy six independent tests against artifact origin:

1. Shell radii do not coincide with disk dynamical scales ($R_{\text{disk}}, 2.15 R_{\text{disk}}, R_{\text{eff}}, r$ at $V_{\text{rot}}^{\text{peak}}$).
2. Shell selections survive 0.1-dex log-normal perturbations to disk and bulge mass-to-light ratios with 86.2% per-fit and 95.1% per-galaxy mode-match against canonical.
3. Shell selections survive distance perturbations drawn from per-galaxy SPARC uncertainties (e_D/D median 8.7%, max 132%) with 89.6% per-fit and 94.1% per-galaxy mode-match.
4. Shell selections survive inclination perturbations drawn from per-galaxy SPARC uncertainties (e_i median 3° , max 10°) with 95.7% per-fit and 98.9% per-galaxy mode-match.
5. Headline structural patterns (mass-radius scaling, width-radius scaling, σ/r universality, morphology gradient, bulge correlation) survive on a conservative anti-warp clean subsample retaining only inner shells in disk-dominated, high-quality, inner-disk regimes.
6. Shell-selection statistics are inconsistent with random or shape-randomized data at the 3σ – 8σ level (§3.2).

No single test is decisive in isolation. Each addresses a specific potential artifact channel: disk dynamics, baryonic input systematics (Υ, D, i), warp geometry, and random structure. The signal survives all six tests in concert. We therefore conclude that the localized structures detected by v7.0 in SPARC galaxies are not consistent with artifacts of the fitting procedure or measurement systematics within the channels tested. Whether they represent genuine localized halo mass features or some other form of organized galactic dynamics remains an open physical question.

4. Discussion

The results presented in §3 establish that the shell population identified by the v7.0 framework exhibits multiple forms of organized statistical structure. These include morphology dependence, correlations with bulge presence, radial scaling relations in shell width and shell mass, coherence-sensitive null-test behavior, and substantial stability under perturbations to the principal baryonic input parameters. Taken

together, these patterns indicate that the shell population is not behaving like an arbitrary collection of unrelated fit artifacts.

At the same time, the present work does not establish a unique physical interpretation for the detected structures. The analysis demonstrates organization and robustness, but does not prove that the localized residual components correspond to discrete physical halo substructures. Several alternative interpretations remain possible, including correlated residual structure arising from non-axisymmetric dynamics, incompletely modeled baryonic effects, or limitations of the adopted smooth-halo backbone. The present paper therefore constrains classes of explanation without uniquely identifying a mechanism.

The spatial-coherence null tests of §3.2 provide the strongest evidence that the shell population depends on organized structure present in the observed rotation curves themselves. The two destructive null procedures fail in opposite directions: residual scrambling suppresses the morphology gradient, while full cross-radius velocity permutation over-produces shell detections. This asymmetric behavior is difficult to reconcile with the hypothesis that the observed shell population arises primarily from random fitting flexibility. In particular, the results imply that localized residual coherence within individual galaxies is essential to reproducing the observed population-level organization.

The perturbation tests of §3.3 further constrain several important artifact channels. Shell selections remain substantially stable under realistic perturbations to stellar mass-to-light ratio, galaxy distance, and inclination, with only modest shifts in population fractions and shell positions. Similarly, the principal organizational signatures persist within the conservatively defined anti-warp clean subsample, suggesting that outer-disk warp artifacts are unlikely to dominate the observed patterns. These results do not eliminate all possible systematic effects, but they substantially narrow the set of simple explanations capable of reproducing the observed population structure.

Several limitations remain important. First, the present analysis depends on the v7.0 decomposition framework and does not establish decomposition uniqueness. Other flexible residual parameterizations or generalized smooth-halo families may potentially absorb part of the observed structure. In particular, comparisons against generalized NFW, free-shape, or hierarchical nonparametric backbones remain necessary future tests. Second, the present work analyzes one-dimensional rotation-curve structure rather than full two-dimensional velocity fields. Explicit modeling of noncircular motions, bars, spiral structure, and warped geometries would provide a stronger separation between localized halo structure and baryonic dynamical effects. Third, the shell population remains modest in size, particularly within the two-shell subsample and the anti-warp clean sample, limiting the statistical power of some higher-order comparisons.

Despite these limitations, the observed organization places meaningful constraints on interpretation. If future analyses confirm that similar residual organization persists across broader backbone families, independent galaxy samples, and higher-dimensional kinematic data, then galaxy rotation curves may contain statistically organized structure beyond conventional smooth-halo descriptions. Whether such structure corresponds to localized dark-matter features, baryon-coupled dynamical phenomena, or another form of organized galactic behavior remains an open question.

Future work should therefore focus on four directions: (i) testing the persistence of the shell population under broader halo-profile families and nonparametric decompositions; (ii) incorporating full velocity-field information from IFU and resolved HI observations; (iii) constructing hierarchical population-level statistical models for shell occurrence and scaling behavior; and (iv) evaluating whether existing dynamical or cosmological simulations naturally reproduce the observed organizational patterns.

5. Conclusions

We have investigated whether the localized residual structures identified in the v7.0 SPARC rotation-curve framework of Paper I behave like a statistically organized population or like an accumulation of unrelated fitting artifacts. Using 102 SPARC galaxies spanning morphological types $T = 2-9$, we analyzed the internal organization, coherence dependence, and robustness properties of the shell-bearing population identified through Bayesian model selection.

The shell population exhibits several forms of empirical organization. Shell-bearing fraction decreases systematically toward later morphological type, while bulge-dominated galaxies are preferentially shell-bearing relative to bulgeless systems. Shell width scales approximately linearly with shell radius, producing a characteristic fractional width $\sigma/r \approx 0.27$ across more than two decades in radius. Within two-shell galaxies, outer shells are systematically more massive than inner shells. These organizational signatures extend the morphology dependence identified in Paper I from a between-galaxy property to a property of the shell population itself.

Two destructive null procedures demonstrate that the morphology gradient depends on coherent residual structure within the observed rotation curves. Residual scrambling suppresses the observed gradient, while cross-radius velocity permutation overproduces shell detections, yielding opposite failure modes relative to the real data. Additional perturbation tests show that shell selections remain substantially stable under realistic variations in stellar mass-to-light ratio, galaxy distance, and inclination. The principal scaling relations also persist within a conservatively defined anti-warp clean subsample.

These results disfavor random fitting behavior, baryonic measurement systematics, and simple warp-dominated explanations as dominant origins of the shell population. However, the present analysis does not establish a unique physical interpretation for the detected structures. The results demonstrate organized residual behavior and robustness against several major artifact channels, but broader comparisons against alternative decomposition frameworks and higher-dimensional kinematic analyses remain necessary.

The principal conclusion of the present work is therefore statistical rather than ontological: the localized residual structures identified by the v7.0 framework behave like a coherent organized population rather than a purely random collection of fit artifacts. Determining the physical origin of that organization remains an open problem for future investigation.

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To be drafted.

Bibliography

To be drafted. Will include: - Bibb (2026) — Paper I - Lelli, McGaugh, & Schombert (2016) — SPARC catalog - Burkert (1995) — Burkert profile - Schwarzschild (1978) — BIC - Relevant rotation-curve substructure and dark matter halo references

Appendix: drafting notes

Sample and convention conventions used throughout

- **102 galaxies** = full v7.0 sample ($T = 2-9$, after Paper I quality cuts)
- **52 shell-bearing galaxies** = Paper I-aligned (NGC 6674 included)
- **69 shells** = total in the per-shell working set (NGC 6674 contributes 2)
- **17 two-shell galaxies** = used in §3.1.4 paired tests
- **26 anti-warp clean shells** = §3.3.5 conservative subsample
- **Morphology gradient** $\rho_{\text{perT}} = -0.833$ = Paper I-aligned
- **Mass bound:** $5 \times 10^{10} M_{\odot}$, binding for $10/69 = 14.5\%$ of shells

Pending small items

- M-r and σ -r slope intercepts still TBD in §3.1.2 equations. Pure cosmetic — slopes (with SEs) and Spearman ρ are now filled in. Intercepts can be added at LaTeX conversion time.
- Paper I citations are placeholders; resolve once Paper I is published.
- All figure references are placeholders; figure generation is a separate task.
- Bibliography is empty.
- Inclination perturbation excludes 11 edge-on galaxies; cosmetic correction ($< 90^\circ \rightarrow \leq 90^\circ$) is a low-priority cleanup.

Word count

Section	Word count
Abstract	290
§2	~1,750
§3.1	~1,650
§3.2	~1,800
§3.3	~2,650
Drafted total	~8,140

§1, §4, §5 still pending. Estimated final length ~10,500–12,000 words including those.
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Part II — Package Status

From STATUS.md in the package root.

Package Sync Status — halo_shells_paper2

Snapshot date: 2026-05-11 **Package version:** v1.0-draft **Purpose of this file:** Single-source-of-truth for what's in this package at this snapshot, what's complete, and what's pending. Read first.

Manuscript

Section	State	Notes
Abstract	First draft, revised	~250 words, tightened
§1 Introduction	First draft	Field framing, SPARC context, Paper I bridge, scope deferrals, outline
§2 Data and methods	First draft	2.1 sample, 2.2 v7.0 framework, 2.3 partitions, 2.4 methods, 2.5 stats conventions
§3.1 Statistical organization	First draft	3.1.1 bulge, 3.1.2 scaling, 3.1.3 σ/r gradient, 3.1.4 inner-vs-outer, 3.1.5 summary
§3.2 Spatial-coherence nulls	First draft	3.2.1 scramble, 3.2.2 permute, 3.2.3 per-T fractions, 3.2.4 joint
§3.3 Robustness against artifacts	First draft	3.3.1 disk scales, 3.3.2-3.3.4 Y/D/i perturbations, 3.3.5 anti-warp clean, 3.3.6 nulls x-ref, 3.3.7 combined
§4 Discussion	First draft	Constrains classes; flags backbone-shape robustness as future work
§5 Conclusions	First draft	“Statistical rather than ontological” framing
Acknowledgments	Pending	Stub only
Bibliography	Pending	Citation hooks in place; full entries needed
Figures (10 total)	Pending	Placeholders annotated in §3 with data sources

Canonical manuscript source: `source/paper2.md`

A previous `source/paper2.tex` file has been removed from this snapshot. It contained content generated during package scaffolding that did not match the mark-down source. LaTeX conversion will be done from `paper2.md` at submission time with appropriate AASTeX 6.3+ scaffolding.

Data files

All files in `data/` are present and aligned with v7.0/v7.1.0 of Paper I. Status:

File	State
<code>data/antiwarp_per_shell.csv</code>	Present (67 shells, NGC 6674 excluded; see note below)
<code>data/antiwarp_summary.txt</code>	Present (regenerated with NGC 6674 included, 69 shells, matches manuscript)
<code>data/upsilon_perturbation_per_galaxy.csv</code>	Present (2,040 rows)
<code>data/distance_perturbation_per_galaxy.csv</code>	Present (2,040 rows)
<code>data/inclination_perturbation_per_galaxy.csv</code>	Present (1,820 rows; 11 edge-on excluded)
<code>data/nulltest_per_realization.csv</code>	Present (40 rows: 2 nulls × 20 reps)
<code>data/nulltest_per_galaxy.csv</code>	Present (4,080 rows)
<code>data/nulltest_summary.txt</code>	Present
<code>data/sparc_sample123.csv</code>	Present (SPARC catalog metadata)
<code>data/galaxy_classifications.csv</code>	Present (with bulge/dwarf/MW-like flags)
<code>data/nfw_fixedc_fits.csv</code>	Present (NFW reference fits)
<code>data/Rotmod_LTG/</code>	Present (175 SPARC rotmod files)

Note on antiwarp CSV/summary sample-size mismatch: The per-shell CSV currently preserves the NGC 6674-excluded version (67 shells), while the summary text was regenerated with NGC 6674 included (69 shells) to match Paper I conventions and the manuscript’s numbers. Regenerating the CSV with NGC 6674 included is on the small-fixes list.

Not included in this package: Paper I canonical fits CSV (`sparc_T2-T9_canonical_fits.csv`). This file lives in the Paper I repository at <https://github.com/RonBibb/sparc-halo-shells/releases/tag/v7.1.0> and is referenced by Paper II scripts. It is not duplicated here to prevent drift between Paper I v7.1.0 and a copy embedded in this package.

Logs

All four production-run logs are present in `logs/`:

- `logs/antiwarp_log.txt`
- `logs/upsilon_perturbation_log.txt`
- `logs/distance_perturbation_log.txt`
- `logs/inclination_perturbation_log.txt`

Scripts

Script	State	Notes
scripts/antiwarp_subsample.py	Reconstructed from manuscript §3.3.5 spec	Pure-analysis script (no fitting); verify against Mac-side reference at ~/Library/CloudStorage/OneDrive-Personal(2)/.../shell_reality_v2/sc
scripts/upsilon_perturbation.py	NOT WRITTEN	Wrapper for Paper I v7.0 fitter; pending
scripts/distance_perturbation.py	NOT WRITTEN	Wrapper for Paper I v7.0 fitter; pending
scripts/inclination_perturbation.py	NOT WRITTEN	Wrapper for Paper I v7.0 fitter; pending
scripts/shell_reality_nulls.py	NOT WRITTEN	Implements scramble/permute nulls; pending

The four wrapper scripts depend on Paper I's `run_canonical_fits.py` as the underlying fitter. For reproduction purposes, the canonical version of each producer script lives on the Mac-side working repository. These scripts can be either ported from the Mac side or reconstructed here as a future task.

Documentation

File	Purpose
STATUS.md	This file — single-source-of-truth for package state
README.md	Package overview, structure, reproduction guide
VALIDATION_STATUS.md	Numerical-claims-to-source mapping; validation tracking
DATA_PROVENANCE.md	Per-CSV producer/inputs/schema/citation trail
ALIGNMENT_AUDIT.md	Paper I <-> Paper II numerical alignment audit
PROJECT_README.md	Original working manifest (preserved)
LICENSE	MIT

File	Purpose
requirements.txt	Python deps for the scripts

What's needed before AJ submission

In rough priority order:

1. **Finish the manuscript text:** acknowledgments paragraph, full bibliography entries with DOIs.
 2. **Generate the 10 figures** from the annotated placeholders in §3. Data is present in data/.
 3. **Port or rewrite the four missing producer scripts** so the package is reproducible from inputs.
 4. **Regenerate antiwarp_per_shell.csv** with NGC 6674 included, so the CSV matches the summary and manuscript.
 5. **Citation hooks -> entries:** the §1 references (Lelli/McGaugh/Schombert, Vegetti, Hezaveh, Gilman, Di Cintio, etc.) need to resolve to actual bibliography entries.
 6. **LaTeX conversion:** convert paper2.md to AASTeX 6.3+ source for submission.
 7. **Update Paper I citation** from “in press” to final volume/page/DOI once Paper I is published.
-

What's deliberately not in this snapshot

- Paper I canonical fits CSV — external, see Paper I repo
 - Paper I production fitter (run_canonical_fits.py) — external
 - Generated figure files (PDFs/PNGs) — to be produced from data + scripts
 - LaTeX manuscript source — to be generated from markdown at submission
 - BibTeX file — to be created at submission
-

Sync verification

If you (Ron) want to verify what's in this package matches what we discussed in conversation, the manuscript content in source/paper2.md should match the markdown you've been editing in this session, including:

- Revised abstract (~250 words, scaling claim toned down, perturbation/null numerics in §3.3/§3.2 only)

- §1 with 7 paragraphs ending in the section outline
- §3.1.3 with the “fractional radial extent” phrasing (no sphere/ring interpretation)
- §4 with 7 paragraphs ending in the four-direction future-work list
- §5 with 5 paragraphs ending in “statistical rather than ontological”
- All 10 figure placeholders in §3 with data-source annotations

If any of those don’t match, the package is out of sync and we should rebuild.

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Part III — Appendix A: Numerical Claims Reference

From VALIDATION_STATUS.md.

Validation Status — halo_shells_paper2 v1.0

Package version: v1.0 **Release date:** 2026-05-10 **Manuscript file:** source/paper2.tex (LaTeX) and source/paper2.md (markdown source) **Parent paper alignment:** Paper I v7.1.0 (<https://github.com/RonBibb/sparc-halo-shells/releases/tag/v7.1.0>)

Version alignment with Paper I

This package is built against Paper I v7.1.0. All canonical fits, framework parameters, and headline numbers are aligned with that release. Specifically:

- **Canonical sample:** 102 SPARC galaxies, $T = 2-9$, after Paper I quality cuts (matches Paper I §2.1)
- **Framework:** v7.0 Burkert backbone + 0/1/2 BIC-selected Gaussian shells, $\sigma/r \leq 0.4$ strict reparameterization, mass bound $5 \times 10^{10} M_{\odot}$ (matches Paper I §2.2)
- **Shell-bearing classification:** 52 of 102 galaxies, including NGC 6674 as shell-bearing (matches Paper I)
- **Morphology gradient:** $\rho_{\text{per } T} = -0.833$, $p = 0.010$ (matches Paper I Table 4)
- **Per-galaxy permutation p:** 0.002 (matches Paper I)

What this v1.0 release contains and what it does not

Included (status: validated against Paper I v7.1.0)

Component	Status
antiwarp_per_shell.csv (67 shells; NGC 6674 excluded as degenerate)	[DONE]

Component	Status
antiwarp_summary.txt (rerun with NGC 6674 included -> 69 shells; matches manuscript table 3.3.5)	[DONE]
upsilon_perturbation_per_galaxy.csv (2,040 rows: 102×20)	[DONE]
distance_perturbation_per_galaxy.csv (2,040 rows)	[DONE]
inclination_perturbation_per_galaxy.csv (1,820 rows; 11 edge-on excluded)	[DONE]
nulltest_per_realization.csv (40 rows: 2×20)	[DONE]
nulltest_per_galaxy.csv (4,080 rows: $2 \times 20 \times 102$)	[DONE]
nulltest_summary.txt	[DONE]
Run logs (antiwarp + 3 perturbations)	[DONE]
SPARC Rotmod files (175 galaxies; 102 used)	[DONE]
Catalog metadata (sparc_sample123.csv, galaxy_classifications.csv)	[DONE]
Manuscript markdown (paper2.md)	[DONE] — drafted ~11,400 words
Manuscript LaTeX (paper2.tex)	[DRAFT] — compiles, but figure includes are stubs

Not yet included or pending external action

Component	Status
Paper I canonical fits CSV	[EXTERNAL] — see Paper I repo at v7.1.0, data/sparc_T2-T9_canonical_fits.csv
Producer scripts	[RECONSTRUCTED] — see notes below
Figures	[pending] 10 placeholders; generation scripts to be written
MD5SUMS for Paper II outputs	[TODO] at finalization
Bibliography full entries	[pending] Skeletons in paper2.tex; verify volume/page/DOI at submission
Validation pass A (numerical claims <-> data)	[TODO]
Validation pass B (cross-section consistency)	[TODO]

Component	Status
Zenodo deposit	[pending] Paper I publication
AJ submission	[pending]

About the reconstructed scripts

The five producer scripts in `scripts/` (`antiwarp_subsample.py`, `upsilon_perturbation.py`, `distance_perturbation.py`, `inclination_perturbation.py`, `shell_reality_nulls.py`) were reconstructed from algorithm specifications in the manuscript and from the run logs in `logs/`. They are not byte-identical to the scripts that produced the v1.0 outputs in `data/`.

Before treating them as canonical: verify the reconstructed scripts against the Mac-side reference scripts at `~/Library/CloudStorage/OneDrive-Personal(2)/Documents/Academi` and reconcile any differences. The most likely sources of drift are: (a) the random seed convention across runs, (b) the exact perturbation distribution form, (c) the exact subprocess-and-pipe-back pattern for invoking `run_canonical_fits.py`. The scientific content (what is computed) should match; the implementation details may differ.

A future revision (v1.1 or later) should replace the reconstructed scripts with the actual Mac-side scripts under MD5 verification.

Headline numerical claims and their data sources

Manuscript claim	Numerical value	Source CSV	Section
Sample size	102 galaxies, $T = 2-9$	<code>sparc_sample123.csv</code> (filtered)	§2.1
Shell-bearing rate	$52/102 = 51.0\%$	derived from Paper I canonical CSV	§3.1
Morphology gradient ρ_{per_T}	-0.833	derived; cross-ref Paper I	§3.1, Abstract
Morphology gradient p	0.010	<code>nulltest_summary.txt</code>	§3.1
Per-galaxy ρ	-0.296	<code>nulltest_summary.txt</code>	§3.1
Per-galaxy p (permutation)	0.002	from Paper I VALIDATION_STATUS	§1, §3.2
M-r slope	0.76 ± 0.14	<code>antiwarp_per_shell.csv</code> (full sample)	§3.1.2
σ -r slope	1.04 ± 0.10	<code>antiwarp_per_shell.csv</code> (full sample)	§3.1.2

Manuscript claim	Numerical value	Source CSV	Section
σ/r median	0.268	antiwarp_per_shell.csv	§3.1.2
σ/r quartile gradient (Q1, Q4)	0.354, 0.185	derived from anti- warp_per_shell.csv	§3.1.3
Bulge OR	3.88 (p = 0.003)	derived from galaxy_classifications + canonical	§3.1.1
Inner-vs-outer mass Wilcoxon	14/17, p = 0.017	derived from anti- warp_per_shell.csv (n=2 paired)	§3.1.4
Inner-vs-outer σ Wilcoxon	12/17, p = 0.049	derived from anti- warp_per_shell.csv	§3.1.4
Two-pop r/r_{vir} KS	D = 0.47, p = 0.045	derived from antiwarp_per_shell + r_{vir}	§3.1.4
Scramble z ($\rho_{\text{per_T}}$)	-4.36	nulltest_summary.txt	§3.2.1
Scramble z ($\rho_{\text{per_galaxy}}$)	-3.04	nulltest_summary.txt	§3.2.1
Permute z ($\rho_{\text{per_T}}$)	-5.21	nulltest_summary.txt	§3.2.2
Permute z ($\rho_{\text{per_galaxy}}$)	-8.33	nulltest_summary.txt	§3.2.2
Y perturbation per-galaxy match	95.1%	upsilon_perturbation_per_galaxy.csv	§3.3.2
Distance perturbation per-galaxy match	94.1%	distance_perturbation_per_galaxy.csv	§3.3.3
Inclination perturbation per-galaxy match	98.9%	inclination_perturbation_per_galaxy.csv	§3.3.4
Anti-warp clean subsample size	26 of 69 shells	antiwarp_summary.txt	§3.3.5
Mass-bound pegging rate	14.5% (10/69)	derived from anti- warp_per_shell.csv	§3.1.2, §4.6
Bound-clean M-r slope	0.67 ± 0.15	derived from anti- warp_per_shell.csv (n=59)	§3.1.2, §4.6

Sample-size conventions

Several conventions vary across the manuscript and need to remain internally consistent:

- **102 galaxies** = full v7.0 sample ($T = 2-9$, after Paper I quality cuts)
- **52 shell-bearing galaxies** = Paper I-aligned (NGC 6674 included)
- **69 shells** = total in the per-shell working set after the NGC 6674-included anti-warp rerun
- **67 shells** = NGC 6674-excluded version preserved in `antiwarp_per_shell.csv` (NGC 6674 contributes 2 ceiling-pegged shells with degenerate $r_1 = r_2 = 3.12$ kpc fit)
- **17 two-shell galaxies** = used in §3.1.4 paired tests (NGC 6674-included)
- **26 anti-warp clean shells** = §3.3.5 conservative subsample (NGC 6674-included rerun)
- **91 galaxies $\times$ 20 reps = 1,820 fits** = inclination perturbation (11 edge-on excluded by $0 < \text{Inc} < 90$ check)

The manuscript uses the NGC 6674-included convention throughout for alignment with Paper I. The `antiwarp_per_shell.csv` file in this package preserves the NGC 6674-excluded version; the `antiwarp_summary.txt` was regenerated with NGC 6674 included and matches the manuscript’s reported numbers.

Known minor issues

1. **Inclination perturbation excludes 11 edge-on galaxies** ($\text{Inc} == 90^\circ$). The exclusion is conservative ($\sin 89^\circ = 0.9998 \approx \sin 90^\circ = 1.000$) and including the 11 galaxies would only increase the apparent stability rate. A future revision could change < 90 to ≤ 90 in the script and rerun for cosmetic completeness. Approximately 9 minutes of CPU time; impact on reported numbers is minimal.
2. **`antiwarp_per_shell.csv` vs `antiwarp_summary.txt` sample-size mismatch.** The CSV contains 67 shells (NGC 6674-excluded version preserved as the catalog), while the `summary.txt` was regenerated with NGC 6674 included (69 shells) to match the manuscript. The discrepancy is documented in `DATA_PROVENANCE.md`; future revisions should regenerate the CSV consistently.
3. **§3.3.7 cross-counting.** §3.3.6 cross-references §3.2 as “channel (vi)” of the artifact-channel battery, while §3.3.7 lists “six independent tests” inclusive of §3.2. Either is defensible; cosmetic only.
4. **Slope intercepts in §3.1.2 equations are TBD.** Slopes (with SEs) and Spearman ρ are filled in; intercepts are pure cosmetic and can be added at LaTeX revision time.

Reproducibility verification

To verify the v1.0 outputs are reproducible from the inputs:

```
cd halo_shells_paper2_v1.0/scripts
python3 antiwarp_subsample.py      # produces data/antiwarp_per_shell.csv, antiwarp_summary.txt
python3 shell_reality_nulls.py     # produces data/nulltest_*.csv, nulltest_summary.txt
python3 upsilon_perturbation.py    # produces data/upsilon_perturbation_per_galaxy.csv
python3 distance_perturbation.py   # produces data/distance_perturbation_per_galaxy.csv
python3 inclination_perturbation.py # produces data/inclination_perturbation_per_galaxy.csv
```

After reproduction, the headline numbers in the regenerated CSVs should match the values in the existing summary files to within statistical fluctuation (the perturbation and null tests use random seeds; numerical reproducibility requires fixed seeds; see script docstrings for seed conventions).

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Part IV — Appendix B: Data Provenance

From DATA_PROVENANCE.md.

Data Provenance — halo_shells_paper2 v1.0

This file documents the provenance of every data file in this package: source script, input dependencies, output schema, and the manuscript section(s) that cite it.

Paper II output CSVs (in data/)

antiwarp_per_shell.csv

- **Producer:** scripts/antiwarp_subsample.py
- **Inputs:** Paper I canonical fits CSV, galaxy_classifications.csv, sparc_sample123.csv, Rotmod_LTG/*.dat
- **Schema:** 67 rows × 18 cols. Columns: Galaxy, T, Q, n_total, position (inner/outer), r_sh_kpc, M_sh, sigma_sh_kpc, sigma_over_r, r_over_RHI, V_gas_at_kms, V_disk_at_kms, V_bulge_at_kms, is_disk_dominated, is_inner, is_bulge_dom, is_bulgeless, is_clean
- **Sample convention:** NGC 6674 EXCLUDED (degenerate fit at $r_1 = r_2 = 3.12$ kpc with both masses pegged at the upper bound). The CSV is preserved with this exclusion. The accompanying antiwarp_summary.txt was regenerated with

NGC 6674 INCLUDED to align with Paper I; this discrepancy is documented in VALIDATION_STATUS.md and is on the future-revisions list.

- **Cited in:** §3.3.5 (anti-warp clean subsample), §3.1.2 (mass-bound check), §3.1.4 (inner-vs-outer paired tests)

antiwarp_summary.txt

- **Producer:** scripts/antiwarp_subsample.py (regenerated rerun with NGC 6674 included)
- **Inputs:** Same as antiwarp_per_shell.csv plus Paper I canonical CSV
- **Schema:** Plain text, formatted output
- **Sample convention:** NGC 6674 INCLUDED, 69 shells across 52 galaxies
- **Cited in:** §3.3.5 Table 3.3.5

upsilon_perturbation_per_galaxy.csv

- **Producer:** scripts/upsilon_perturbation.py
- **Inputs:** SPARC Rotmod files, sparc_sample123.csv, Paper I v7.0 production fitter (run_canonical_fits.py from Paper I repo)
- **Test design:** 20 realizations per galaxy. Each draws ($\log Y_{\text{disk}}$, $\log Y_{\text{bulge}}$) from independent normal distributions with means ($\log 0.5$, $\log 0.7$) and standard deviation 0.1 dex. Refits the v7.0 framework on each perturbed input.
- **Schema:** 2,040 rows $\times$ 20 cols (102 galaxies $\times$ 20 realizations; 39 fits failed on $V_{\text{obs}}^2 > V_{\text{bar}}^2$ mask after Y rescaling). Columns: Galaxy, realization, Upsilon_disk, Upsilon_bulge, status, n_shells, n_points, rho0, a_kpc, chi2, chi2_red, bic, r_sh1, M_sh1, sigma_sh1, sigma_over_r1, r_sh2, M_sh2, sigma_sh2, sigma_over_r2
- **Cited in:** §3.3.2; §3.3.7 channel (ii)
- **Run log:** logs/upsilon_perturbation_log.txt
- **Headline result:** Per-galaxy modal n_shells matches canonical in 95.1% of galaxies; per-fit match 86.2%.

distance_perturbation_per_galaxy.csv

- **Producer:** scripts/distance_perturbation.py
- **Inputs:** Same as upsilon_perturbation
- **Test design:** 20 realizations. Each draws D' from $N(D, e_D)$ using SPARC catalog uncertainties, refits with perturbed radii ($r \rightarrow r \cdot D'/D$) and perturbed baryonic velocities ($V_{\text{bar}} \rightarrow V_{\text{bar}} \cdot \sqrt{D'/D}$). V_{obs} unchanged.
- **Schema:** 2,040 rows $\times$ 22 cols. Columns: Galaxy, realization, D_nominal_Mpc, D_perturbed_Mpc, distance_factor, e_D_Mpc, status, n_shells, n_points, rho0, a_kpc, chi2, chi2_red, bic, r_sh1, M_sh1, sigma_sh1, sigma_over_r1, r_sh2,

M_sh2, sigma_sh2, sigma_over_r2

- **Cited in:** §3.3.3; §3.3.7 channel (iii)
- **Run log:** logs/distance_perturbation_log.txt
- **Headline result:** Per-galaxy modal match 94.1%; per-fit 89.6%. Fractional perturbation magnitudes: median 8.7%, mean 13.8%.

inclination_perturbation_per_galaxy.csv

- **Producer:** scripts/inclination_perturbation.py
- **Inputs:** Same as uppsilon_perturbation
- **Test design:** 20 realizations. Each draws Inc' from $N(\text{Inc}, e_{\text{Inc}})$ per-galaxy, rescales V_obs and e_V_obs by $\sin(\text{Inc})/\sin(\text{Inc}')$. 11 galaxies at Inc = 90° excluded by $0 < \text{Inc} < 90$ check.
- **Schema:** 1,820 rows × 22 cols (91 galaxies × 20 realizations). Columns include Inc_nominal_deg, Inc_perturbed_deg, e_Inc_deg, vobs_factor, plus the same fit columns as the other perturbation CSVs.
- **Cited in:** §3.3.4; §3.3.7 channel (iv)
- **Run log:** logs/inclination_perturbation_log.txt
- **Headline result:** Per-galaxy modal match 98.9%; per-fit 95.7%. Tightest position recovery of the three perturbation tests (median $\Delta \log r_1 = 0.005$ dex).

nulltest_per_realization.csv

- **Producer:** scripts/shell_reality_nulls.py
- **Inputs:** SPARC Rotmod files, Paper I canonical CSV, v7.0 production fitter
- **Test design:** Two null types — *scramble* (within-galaxy permutation of dark-matter residuals around the canonical Burkert backbone) and *permute* (within-galaxy permutation of V_obs values across radii). 20 realizations per null type. Each realization fits all 102 galaxies and computes per-T-bin and per-galaxy Spearman ρ .
- **Schema:** 40 rows × 24 cols (2 null types × 20 realizations). Columns: null_type, realization, rho_per_T, p_per_T, rho_per_gal, p_per_gal, n_shellbearing, n_total, plus per-T shell-bearing fractions (frac_T2 through frac_T9 with their counts).
- **Cited in:** §3.2

nulltest_per_galaxy.csv

- **Producer:** scripts/shell_reality_nulls.py
- **Inputs:** Same as nulltest_per_realization.csv
- **Schema:** 4,080 rows × 8 cols (2 null types × 20 realizations × 102 galaxies). Columns: null_type, realization, Galaxy, T, n_pts, n_shells, chi2, bic.
- **Cited in:** §3.2 (per-galaxy fit details available for stratification)

nulltest_summary.txt

- **Producer:** scripts/shell_reality_nulls.py (formatted output)
 - **Schema:** Plain text formatted summary
 - **Cited in:** §3.2.1, §3.2.2, §3.2.3, §3.2.4
-

Inputs from external sources (in data/)**sparc_sample123.csv**

- **Source:** SPARC catalog [Lelli, McGaugh, Schombert 2016, AJ 152, 157]
- **Schema:** 123 rows $\times$ 25 cols. Standard SPARC catalog columns plus derived M_{star} , r_{vir} , M_{halo} via abundance matching (procedure described in Paper I §2.1).
- **Cited in:** §2.1 (sample), §2.2 (catalog metadata for fitting)

galaxy_classifications.csv

- **Source:** Augmented from sparc_sample123.csv with bulge/dwarf/MW-like classifications derived from the SPARC photometric decomposition.
- **Schema:** 123 rows $\times$ 32 cols. sparc_sample123 columns plus 7 boolean flags: is_dwarf, is_mw_like, is_bulge_dom, is_bulgeless, is_transitional, max_bulge_frac, has_bulge_col.
- **Cited in:** §3.1.1 (bulge correlation)

nfw_fixedc_fits.csv

- **Source:** Paper II reference; NFW fixed-concentration fits using the DM14 prior.
- **Schema:** 123 rows $\times$ 12 cols.
- **Cited in:** §1 (introduction context); not the primary fitting framework of the paper.

Rotmod_LTG/

- **Source:** SPARC distribution. 175 SPARC galaxies; 102 are used in the $T = 2-9$ sample after quality cuts.
 - **Schema:** Tab-separated ASCII files with columns: Rad (kpc), V_{obs} (km/s), $\text{err}V$ (km/s), V_{gas} , V_{disk} , V_{bulge} , SBdisk, SBbul.
 - **Cited in:** §2.1; primary input to all fitting analyses.
-

Inputs that must be supplied externally

Paper I canonical fits CSV (sparc_T2-T9_canonical_fits.csv)

- **Source:** Paper I repository at v7.1.0, in data/
- **Required for:** All analyses in scripts/ that reference per-galaxy shell parameters
- **Why not duplicated here:** To prevent drift between Paper I v7.1.0 and a copy embedded in the Paper II repository. The Paper I repo is the canonical source.
- **How to obtain:**

```
git clone https://github.com/RonBibb/sparc-halo-shells.git
cd sparc-halo-shells && git checkout v7.1.0
cp data/sparc_T2-T9_canonical_fits.csv ../halo_shells_paper2_v1.0/data/
```

Paper I v7.0 production fitter (run_canonical_fits.py)

- **Source:** Paper I repository at v7.1.0, in scripts/
- **Required for:** The four perturbation runner scripts (upsilon_, distance_, inclination_, shell_reality_nulls) which call out to the v7.0 fitter on perturbed inputs
- **Why not duplicated:** Paper I is the canonical source for the fitter; embedding a copy risks divergence
- **How to obtain:**

```
cp sparc-halo-shells/scripts/run_canonical_fits.py ../halo_shells_paper2_v1.0/scripts/
```

Section-to-data quick reference

Manuscript section	Primary data	Secondary data
§2.1 Sample	sparc_sample123.csv, galaxy_classifications.csv	Rotmod_LTG/
§2.2 Framework	(Paper I canonical CSV — external)	—
§3.1.1 Bulge correlation	galaxy_classifications.csv + Paper I canonical	antiwarp_per_shell.csv
§3.1.2 Scaling relations	antiwarp_per_shell.csv	antiwarp_summary.txt
§3.1.3 σ/r quartile gradient	antiwarp_per_shell.csv	—

Manuscript section	Primary data	Secondary data
§3.1.4 Inner-vs-outer	antiwarp_per_shell.csv + sparc_sample123.csv (r_vir)	—
§3.2.1 Scramble null	nulltest_per_realization.csv	nulltest_summary.txt
§3.2.2 Permute null	nulltest_per_realization.csv	nulltest_summary.txt
§3.2.3 Per-T fractions	nulltest_summary.txt	nulltest_per_galaxy.csv
§3.3.1 Disk dynamical scales	(analysis on Paper I canonical CSV)	—
§3.3.2 Y perturbation	upsilon_perturbation_per_galaxy.csv	logupsilon_perturbation_log.txt
§3.3.3 Distance perturbation	distance_perturbation_per_galaxy.csv	logdistance_perturbation_log.txt
§3.3.4 Inclination perturbation	inclination_perturbation_per_galaxy.csv	loginclination_perturbation_log.txt
§3.3.5 Anti-warp clean subsample	antiwarp_summary.txt	antiwarp_per_shell.csv
§3.3.6 Cross-ref to §3.2	nulltest_summary.txt	—
§3.3.7 Combined verdict	(citations above)	—